Master's Thesis Presentation:
AI-Empowered Optical IRS-Aided Non-Terrestrial Networks

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Outline of presentation

I. Introduction

II. System model

III. Proposed AI-based Optimization Framework

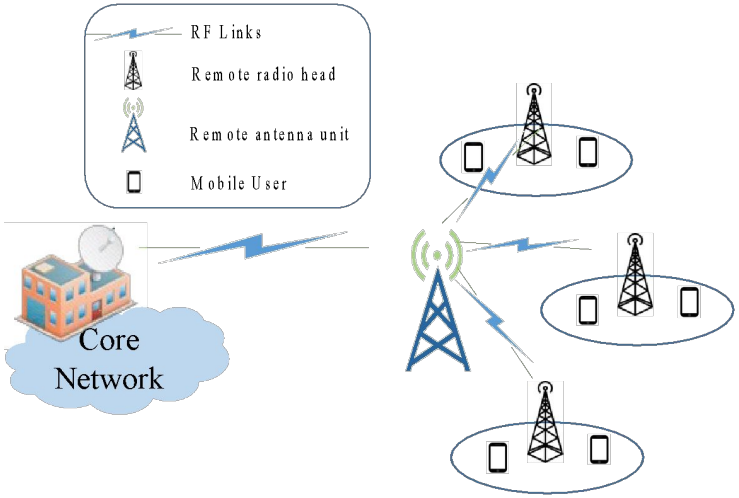
IV. Experimental results

V. Conclusion

I. Introduction

Coverage Issue: Aerial Networks

- The current 5G networks mainly rely on terrestrial networks
  - This limits the coverage area, especially for rural and remote areas.

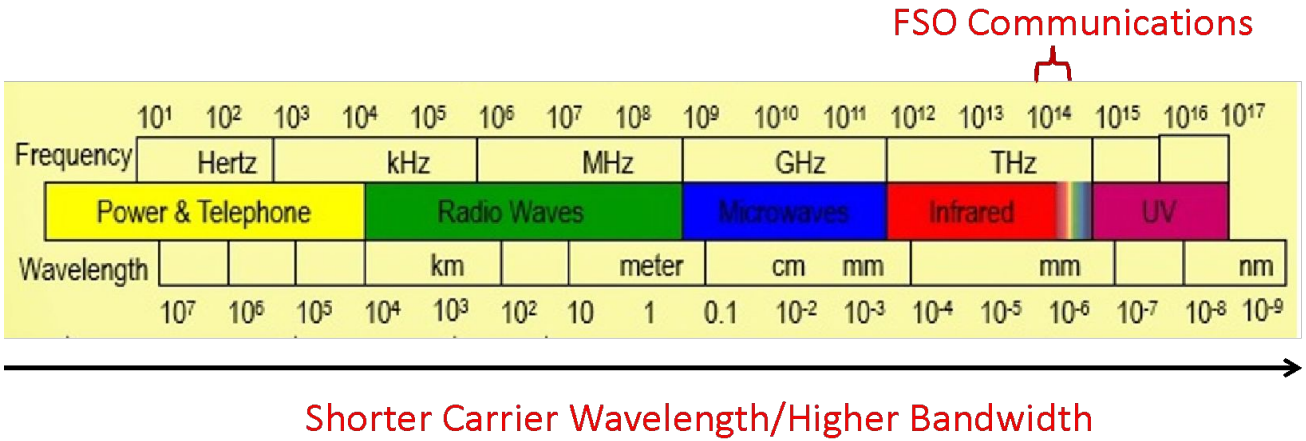


- Possible solution: using *aerial networks*
  - HAP (20 km) and UAV (up to km)
  - Cost-effective solutions
  - Flexible deployment
  - Wide coverage area



High-speed Connections: Free-space Optics

- The current 5G networks use RF communications
  - Spectrum congestion and limited data rate
  - Emerging applications require high data rate. e.g., Gbps up to Tbps
- FSO using infrared bands is a potential solution
  - Large bandwidth → offer high-speed connections (~Gbps or even Tpbs)
  - Robustness to electromagnetic interference
  - Inherent security and unlicensed spectrum

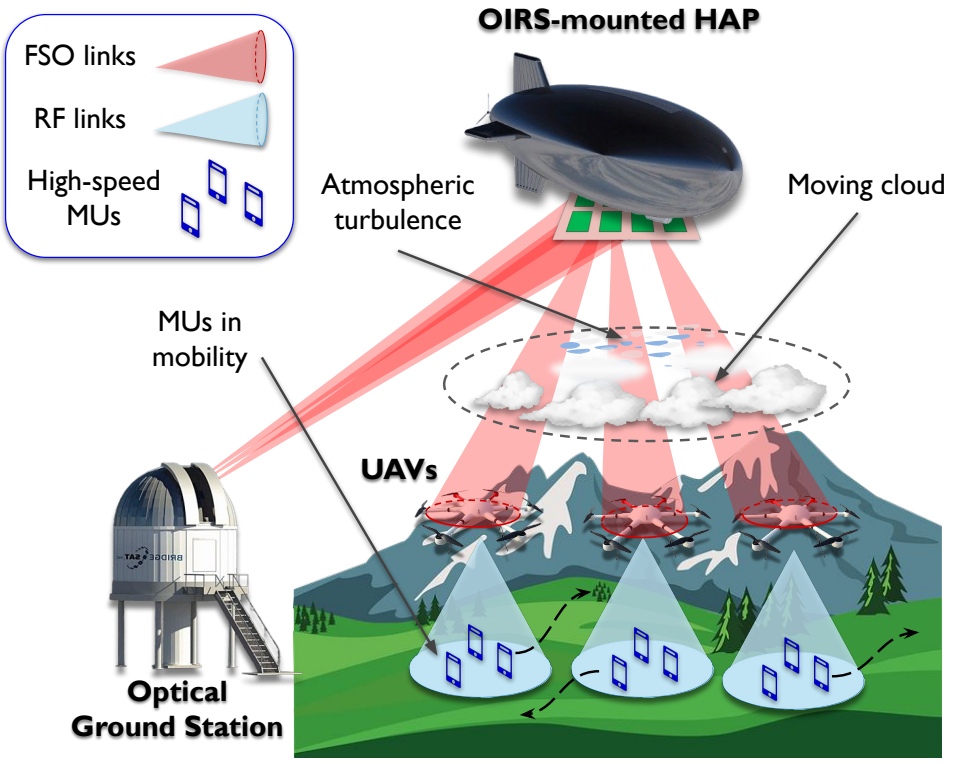


FSO-based Air-ground Integrated Networks

We introduce an air-ground integrated network (AGIN) architecture

→ To deliver high-speed connectivity to hard-to-reach areas and quickly deployable scenarios, such as temporary events

- OIRS-aided HAP-based FSO backhaul
  - HAP (stable at 20 km altitude) providing wide coverage (~50 km)
  - Optical Intelligent Reflecting Surface (OIRS)[*]: low cost and low energy consumption
- UAV-mounted base stations for RF last-mile access
  - Flexible deployment to serve mobile users in hard-to-reach areas and temporary events



[*] OIRS is defined as a "mirror" reflecting the signals to overcome the blockage in LoS wireless communications

Challenging Issue: Dynamic Networks

- Challenging issues in our proposed AGIN network: Dynamic networks on backhaul and access links
  - Backhaul: Moving clouds
  - Access: Mobile users in mobility
- Real-time network optimization under dynamic networks
  - Conventional optimization (e.g., metaheuristics,...): not suitable for real-time implementation due to the high computational time
  - AI-based optimization (e.g. Deep Reinforcement Learning,...): suitable for real-time implementation thanks to the low computational time (e.g. milliseconds) *[1]*

[1] *L. Huang, S. Bi, and Y.-J. A. Zhang, "Deep reinforcement learning for online computation offloading in wireless powered mobile-edge computing networks," IEEE Transactions on Mobile Computing, vol. 19, no. 11, pp. 2581–2593, Nov. 2020.*

→ AI-based optimization (DRL) is more suitable for real-time optimization in dynamic networks

Study Goal: Joint Optimization in Dynamic Networks

Our goal: Find optimal UAV placement and OIRS allocation to serve as many users as possible while saving UAV energy consumption in our introduced AGIN network, considering dynamic both backhaul and access links.



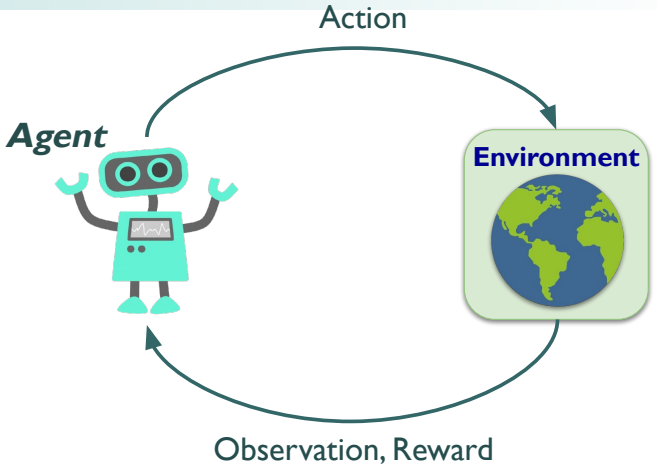
Research question: How to jointly optimize them under the dynamic conditions from both backhaul (e.g., movement of clouds) and last-mile access (e.g., movement of mobile users) networks?



Our approach: Propose an AI-based optimization solution for the joint optimization, given the dynamic backhaul and access networks

AI-based optimization: Deep Reinforcement Learning

Deep Reinforcement Learning (DRL) is the combination of *reinforcement learning* and *deep neural networks*, enabling an agent to learn in an interactive environment by trial and error using feedback from its own experiences

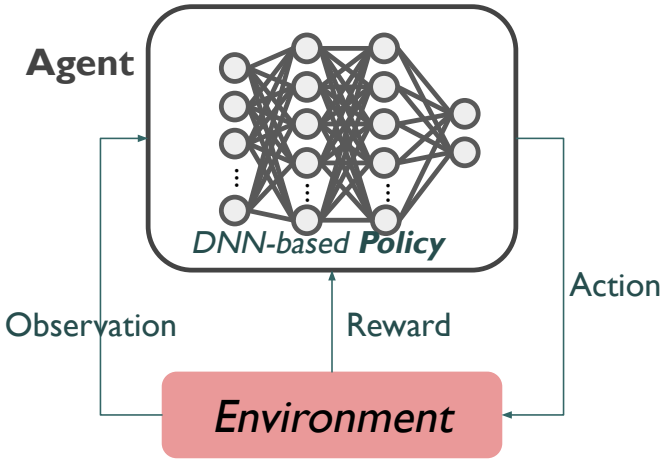


DRL is modeled as follows:

- **Observation:** current environment condition
- **Action:** decision taken by the agent
- **Reward:** feedback signal from Environment
- **Policy:** DNN mapping from states to actions

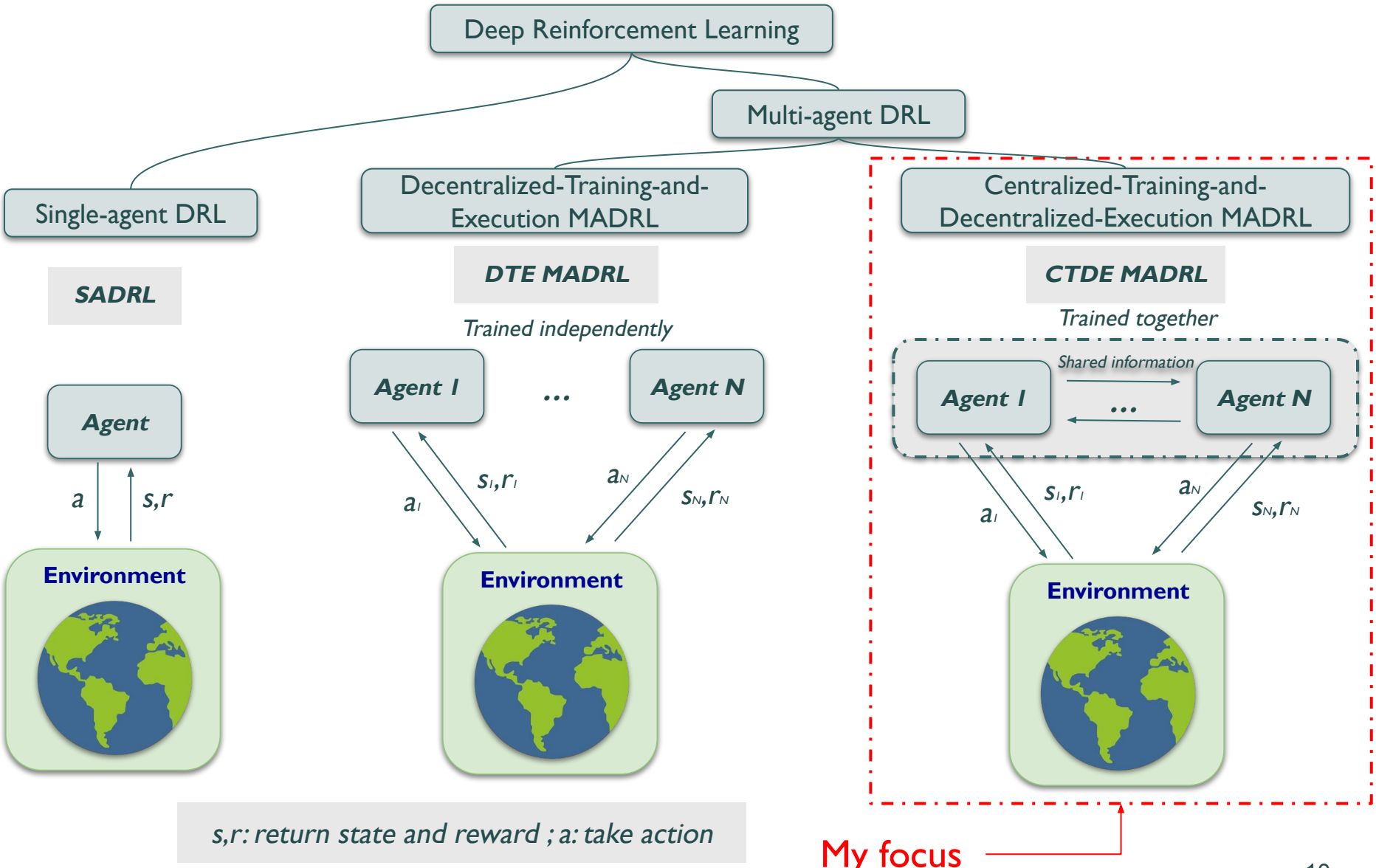
→ ***Agent:*** *Maximize the reward*

Policy(Observation) → Action



My Goal is to jointly control UAVs and OIRS allocation ➡ **Multi agents**

DRL: Single-agent and Multi-agent



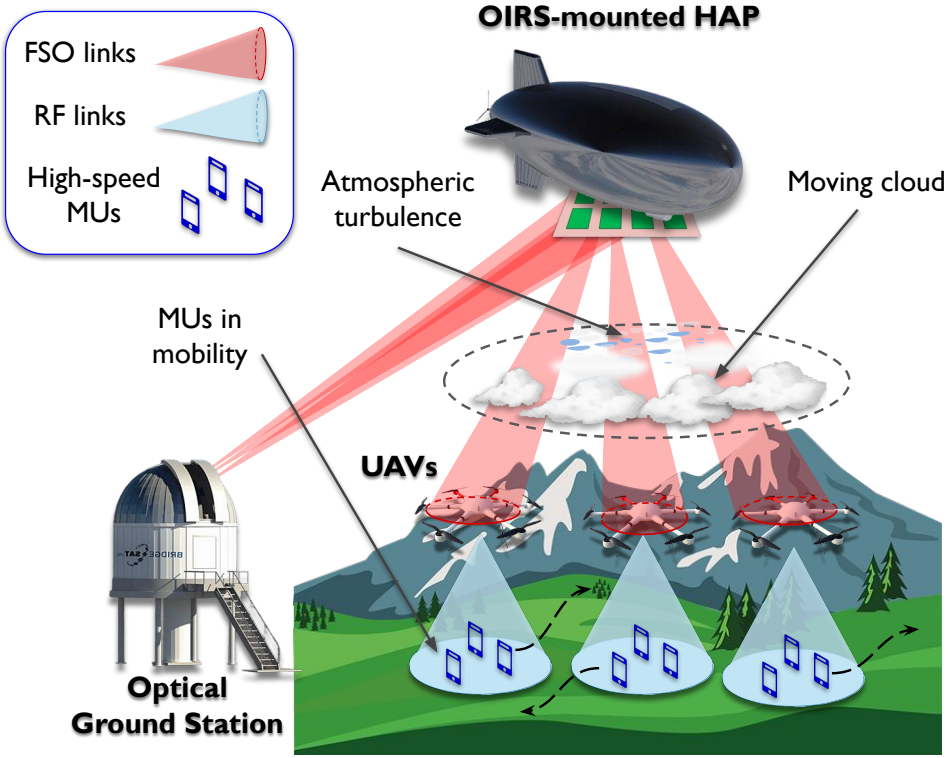
II. System model

System model: Backhaul and access networks

Our introduced AGIN network:

The HAP-based FSO backhaul and UAV-mounted base stations for RF last-mile access are deployed to provide high-speed data rate to users in rural/remote areas or temporary events.

- OIRS-aided HAP-based FSO backhaul:
  - Optical laser signals transmitted by the optical ground station are reflected by the OIRS mounted on the HAP toward the UAVs.
  - Challenge: Moving clouds lead to the fluctuation and constraint of backhaul capacity over time
- UAV-mounted base stations for RF last-mile access:
  - Multiple UAVs are deployed to serve mobile users (MUs).
  - Challenge: MUs in mobility make access network dynamic



Objective: Serve as many MUs as possible while saving UAV energy consumption, given the dynamic conditions of both backhaul and access networks.

Problem Formulation: Joint Optimization

Problem: Find optimal UAV placement and OIRS element allocation to serve as many MUs as possible while keeping the UAV energy consumption low.

Given *M* UAVs are deployed to support *N* MUs over *T* time slots. Our optimization problem is formulated as:

Number of served MUs

UAVs' Energy Consumption

$$\mathcal{P}: \max_{\mathcal{U},\mathcal{A}} \quad Z = \sum_{t=1}^{T}\sum_{m=1}^{M} |\mathcal{H}_{m,t}| - \Lambda \sum_{t=1}^{T}\sum_{m=1}^{M} P(v_{m,t}),$$

$$\text{s.t.} \quad |\mathcal{H}_{m,t}| R_{\text{target}}^{\text{RF}} \le C_{m,t}^{\text{FSO}}, \forall m,$$

The constraint of backhaul capacity

$$\sum_{n \in \mathcal{H}_{m,t}} P_{m,n,t}^{\text{UAV}} \le P_{m,\max}^{\text{UAV}}, \forall m,t$$

The constraint of UAV's communication power

$$x_{\min} \le x_{m,t}^{\text{UAV}} \le x_{\max}, \forall m,t,$$
$$y_{\min} \le y_{m,t}^{\text{UAV}} \le y_{\max}, \forall m,t,$$

Placement constraints of UAVs in the target area

$$v_{m,t} \le v_{\max}^{m}, \forall m,t$$

Maximum velocity of UAVs

$$\sum_{m=1}^{M} \Sigma_{\text{IRS}}^{m,t} = \Sigma_{\text{IRS}}, \forall t$$

The constraint of element allocation

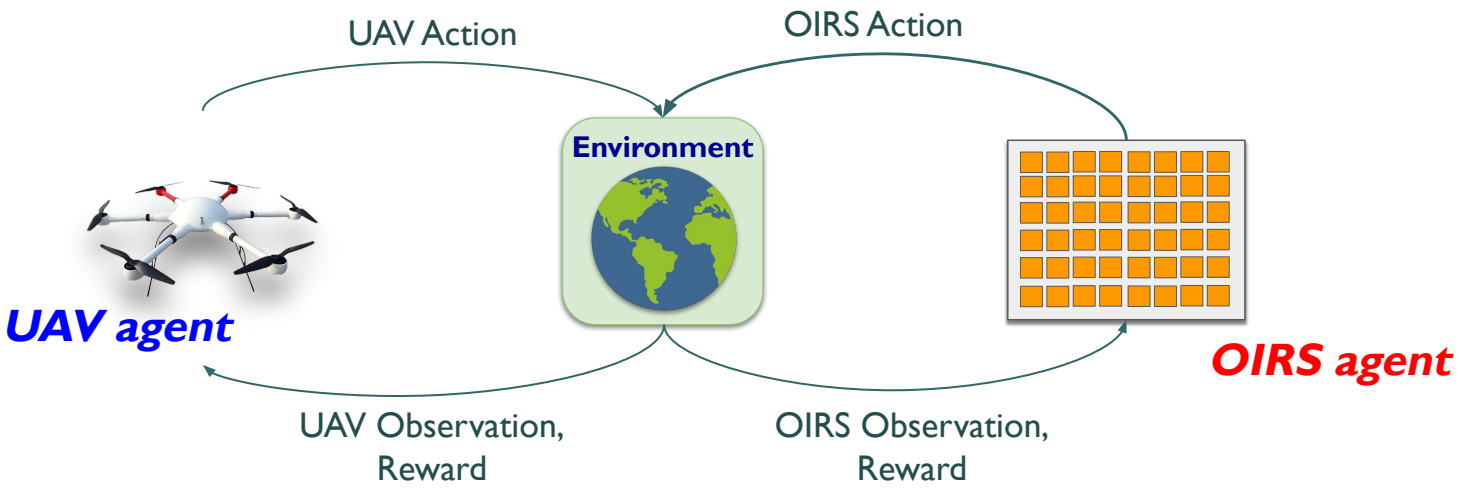
UAV Placement, OIRS element allocation

We apply MADRL to address this optimization problem.

III. Proposed AI-based Optimization Framework

MADRL: UAV agent and OIRS agent

We design 2 different types of agents: ***UAV agent*** and ***OIRS agent***



Observation:

- **m-th UAV State:** [UAVs' locations, Heatmap of unsupported MUs, Heat map of MUs within m-th UAV's coverage, Cloud heatmap, Current backhaul capacity, Total supported MUs]
- **OIRS State:** [UAVs' locations, Heatmap of unsupported MUs,Heat map of MUs within m-th UAV's coverage, Heat map of MUs within each UAV's coverage]

Action: Continuous Control

- **UAV Action:** UAV displacement
- **OIRS Action:** OIRS element allocation probability (e.g. 30% elements for UAV1, 50% for UAV2,...)

Reward:

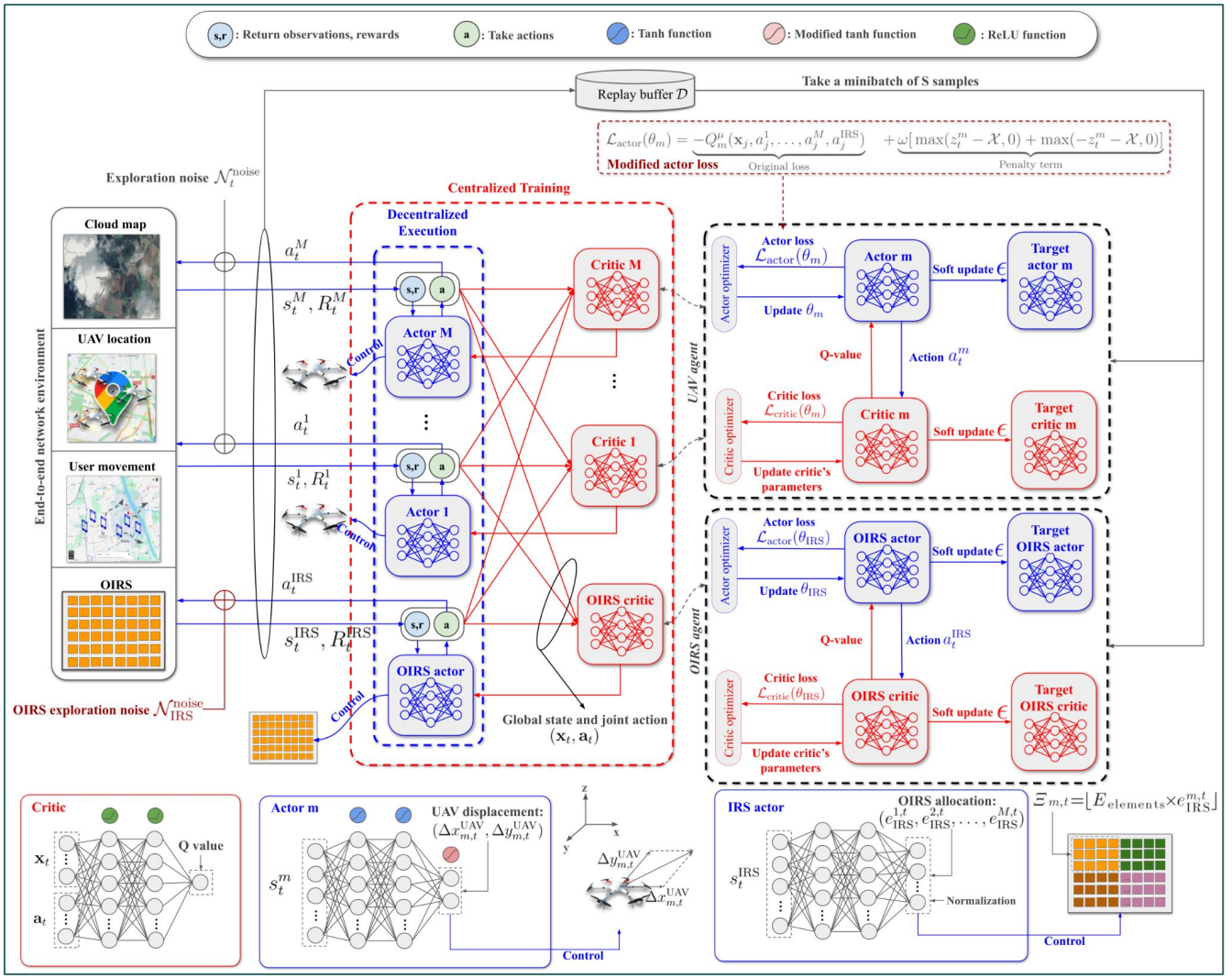
- **UAV Reward:** Based on the number of served MUs and UAV power consumption
- **OIRS Reward:** Based on the number of served MUs and STD of unserved MUs and STD of remaining backhaul.

MADRL-based Framework

Our proposed architecture is built based on MADDPG [1], a CTDE-based MADRL:

Novel components in our proposed architecture:

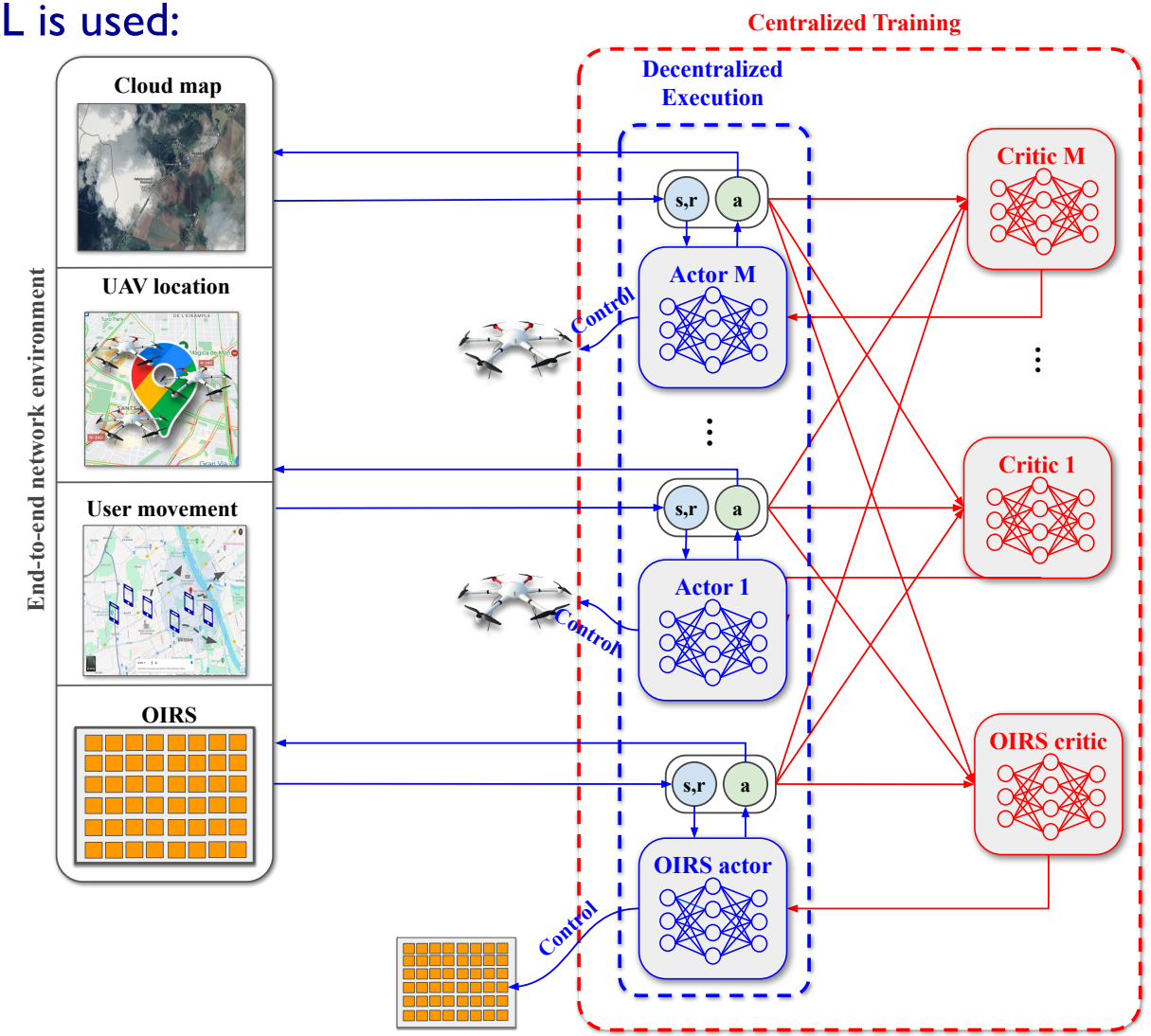
1. *Heterogeneous actor networks* are designed to control UAV placement and OIRS element allocation.
2. Mitigating saturation problem through *modifications to the tanh function, actor loss*.
3. Mitigating invalid actions of OIRS agent through *modifications to exploration noise*.



[1] R. Lowe, Y. Wu, A. Tamar, J. Harb, P. Abbeel, and I. Mordatch, "Multi- agent actor-critic for mixed cooperative competitive environments," Adv. Neural Inf. Process. Syst., vol. 30, 2017.

MADRL-based Framework: CTDE

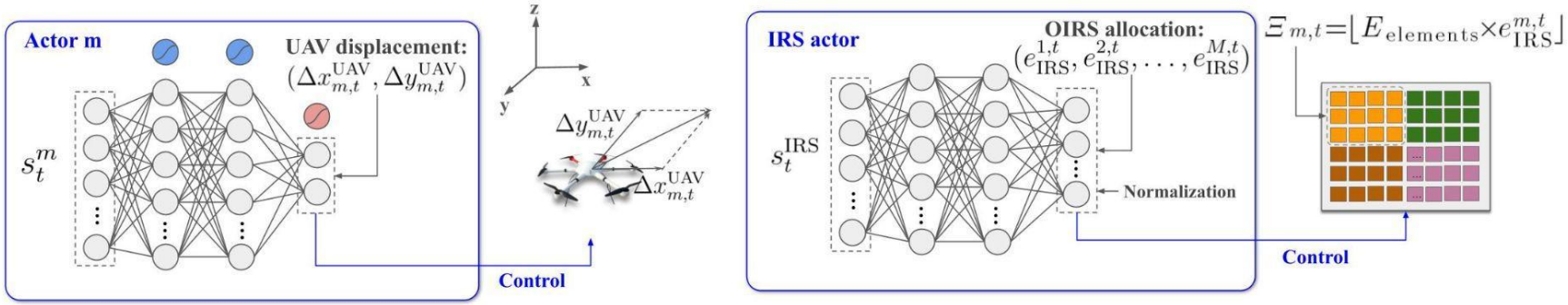
The high cooperation among UAVs' placement and OIRS allocation is required, so a CTDE-based MADRL is used:



MADRL Framework: Heterogeneous Actors

Novel components in our proposed architecture:

1. Heterogeneous actor networks are designed to control UAV placement and OIRS element allocation.

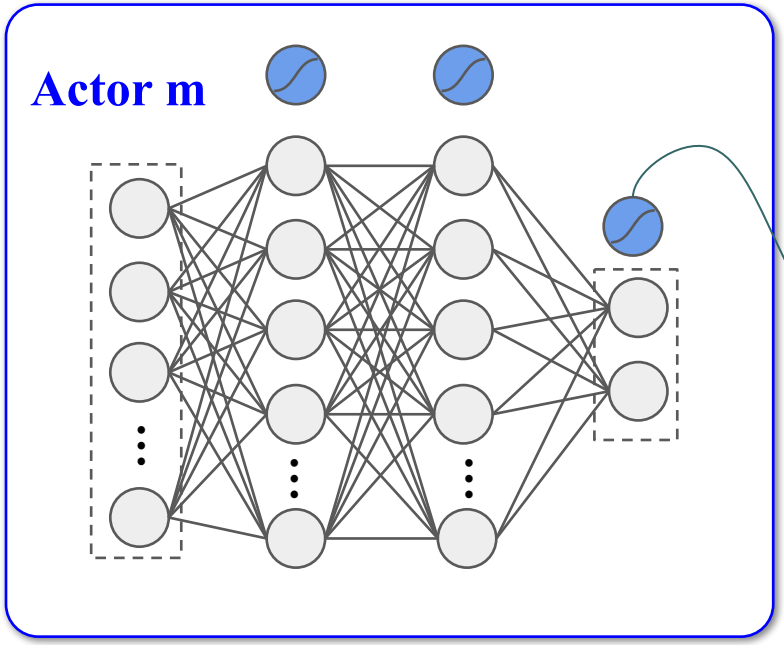


MADRL Framework: Tanh Function & Actor Loss

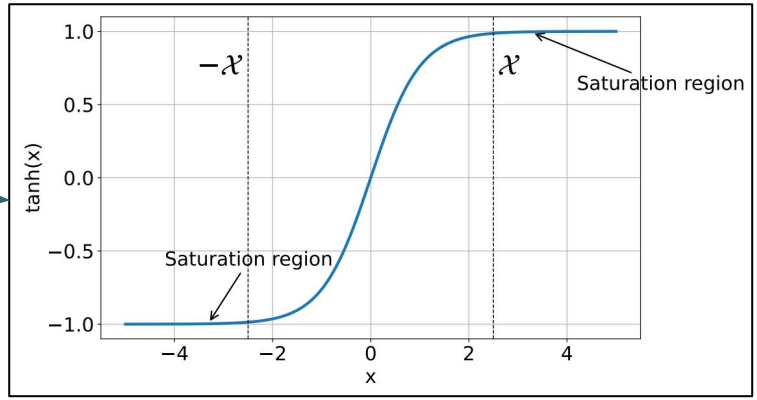
Novel components in our proposed architecture:

2. Mitigating saturation problem [1] through *modifications to the tanh function, actor loss*.

Saturation problem [1] refers to the phenomenon in continuous DRL where the continuous action values output by tanh or sigmoid functions fall into their saturation regions:



The saturation region of tanh function:



[1] *R. Ding, F. Gao, and X. S. Shen, "3D UAV trajectory design and frequency band allocation for energy-efficient and fair communication: A deep reinforcement learning approach," IEEE Trans. Wireless Commun., vol. 19, no. 12, pp. 7796–7809, 2020*

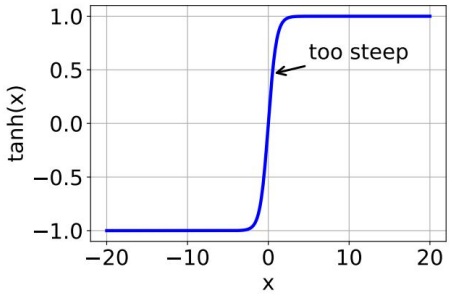
MADRL Framework: Tanh Function & Actor Loss

Novel components in our proposed architecture:

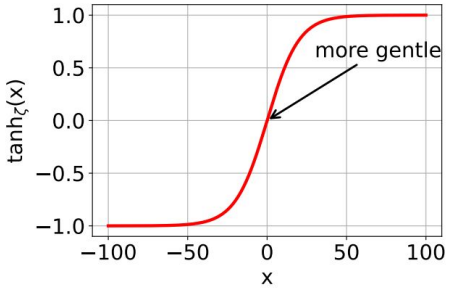
2. Mitigating saturation problem [1] through *modifications to the tanh function, actor loss*.

Original tanh function:

$$\frac{e^{x}-e^{-x}}{e^{x}+e^{-x}}$$



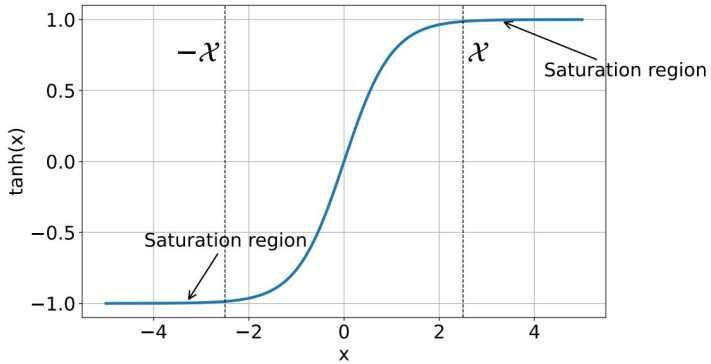
Modified tanh function:



$$\frac{e^{\zeta x}-e^{-\zeta x}}{e^{\zeta x}+e^{-\zeta x}}$$

Modified actor loss:

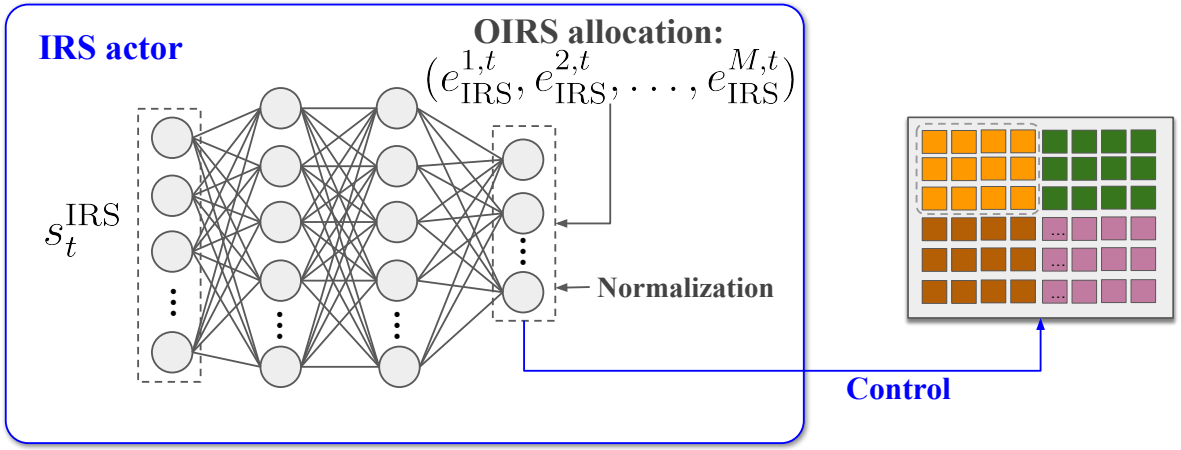
$$\mathcal{L}_{\text{actor}}(\theta_m) = \underbrace{-Q_m^{\mu}(\mathbf{x}_j, a_j^1, \ldots, a_j^M, a_j^{\text{IRS}})}_{\text{Original loss}} + \underbrace{\omega[\max(z_t^m - \mathcal{X}, 0) + \max(-z_t^m - \mathcal{X}, 0)]}_{\text{Penalty term}}$$



MADRL Framework: Invalid Actions

Novel components in our proposed architecture:

3. Mitigating invalid actions of OIRS agent through *modifications to exploration noise*.



Problem of OIRS Action:

$$a_t^{\text{IRS}} = [e_{\text{IRS}}^{1,t}, e_{\text{IRS}}^{2,t}, \ldots, e_{\text{IRS}}^{M,t}]$$

Must satisfy: $\sum_{m=1}^{M} e_{\text{IRS}}^{m,t} = 1$

However, adding exploration noise may violate this constraint

Proposed modification:

→ Exploration noise for OIRS action is redefined as: $\mathcal{N}_{\text{IRS}}^{\text{noise}} = [n_1, n_2, \ldots, n_M]$

Where: $n_M = 1 - (n_1 + n_2 + \cdots + n_{M-1}) \quad n_i \sim \mathcal{N}(\mu, \sigma^2) \forall i \in \{1, 2, \ldots, M-1\}$

→ Always guarantee: $\sum_{m=1}^{M} e_{\text{IRS}}^{m,t} = 1$

IV. Experimental Results

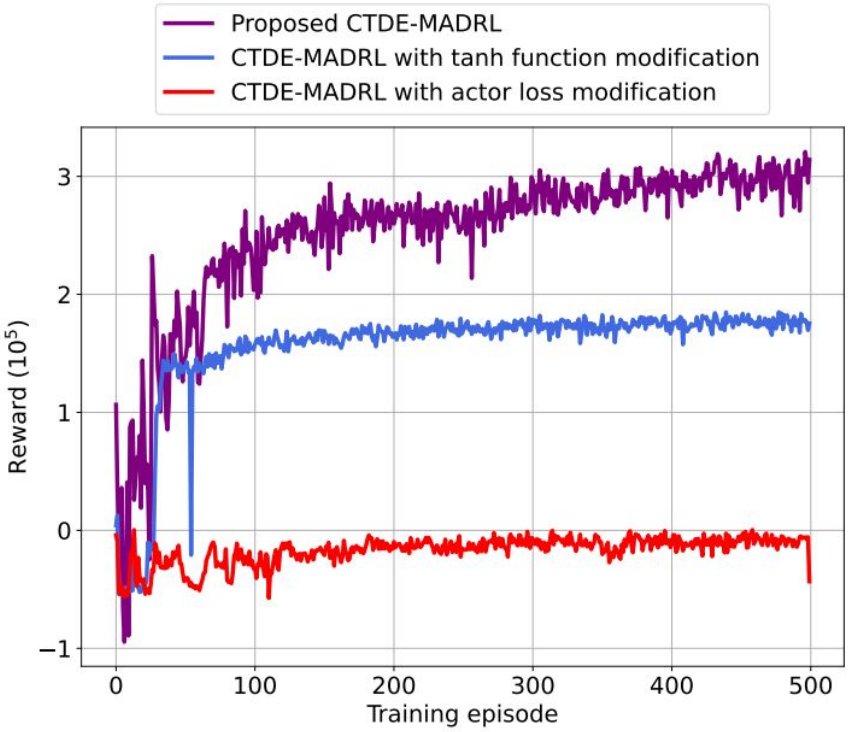
Experimental Results: Parameter Settings

- M = 3 UAVs
- Target area size: 2km×2km
- N = 300 MUs, following a Gaussian distribution
- 20% of the MUs follow the Gauss-Markov mobility

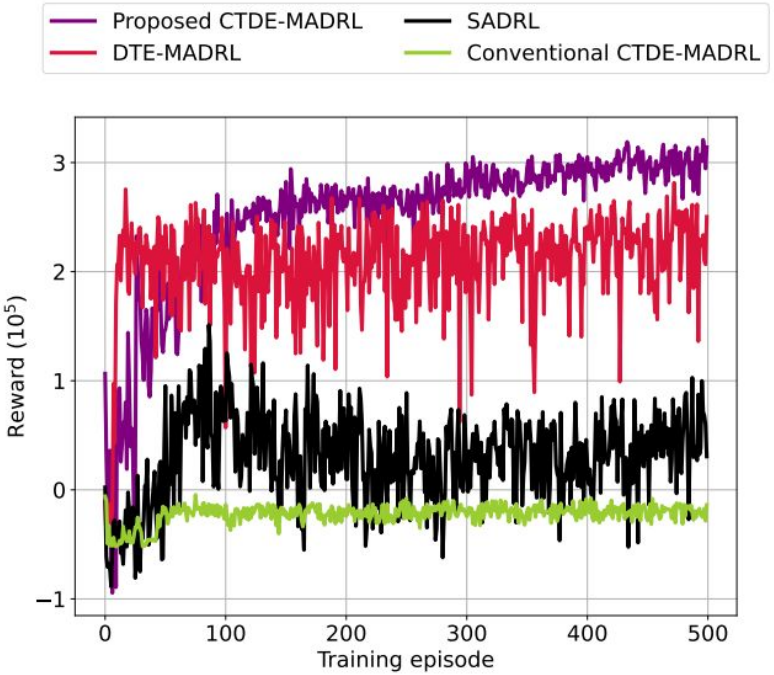
Backhaul	Description	Value
h^{HAP}	HAP's altitude	20 (km)
$d^{u}_{m,t}$	Ground-HAP distance	40 (km)
$P_{m,\text{T}}$	Transmitted power	1 (W)
w^{u}_{o}	Beam waist	0.02(m)
$\lambda_1, \lambda_2, \lambda_3$	Optical wavelengths	1550, 1555, 1560 (nm)
σ^2_{FSO}	Noise power	10^{-5} (W)
$B_{m,\text{FSO}}$	FSO bandwidth	1 (GHz)
E_{elements}	Number of OIRS elements	100
$L_x \times L_y$	OIRS's size	25(cm) × 25 (cm)
a_m	UAV detector radius	10 (cm)
η_m	Detector responsivity	0.9
Access	**Description**	**Value**
h^{UAV}_{m}	UAV's altitude	350 (m)
v^{m}_{max}	UAV's maximum velocity	40 (m/s)
φ_m	UAV's half divergence angle	$\pi/4$ (rad)
B_{RF}	RF bandwidth per MU	20 (MHz)
$f_{1,c}, f_{2,c}, f_{3,c}$	UAVs' carrier frequencies	5,6,7 (Hz)
$P^{\text{UAV}}_{m,\text{max}}$	UAV's max transmit power	10 (W)
α_m	Path loss exponent	2.7
σ^2_{RF}	Noise power	10^{-14} (W)
$R^{\text{RF}}_{\text{target}}$	Targeted data rate of MUs	100 (Mbps)
κ, ψ_L, ψ_M	Rician channel's parameters	50,1,$\mathcal{CN}(0,1)$

Experimental Results: Reward Convergence

Proposed CTDE-based MADRL under the exclusion of each modification:



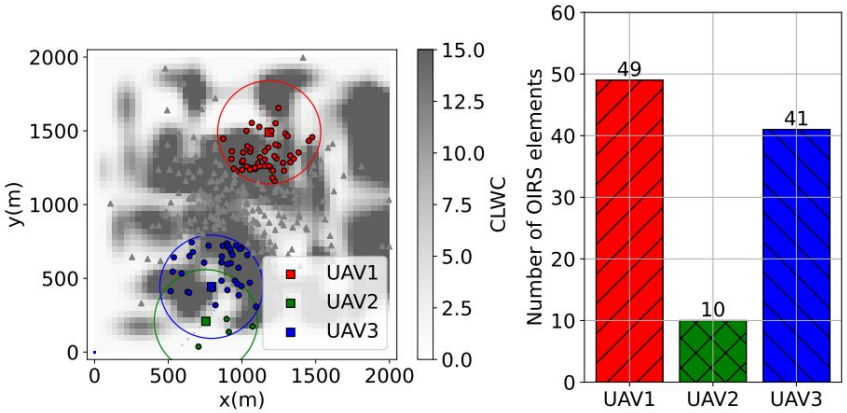
Reward convergence among the proposed MADRL model and benchmark methods:



Experimental Results: The Effectiveness of Adaptive OIRS Allocation

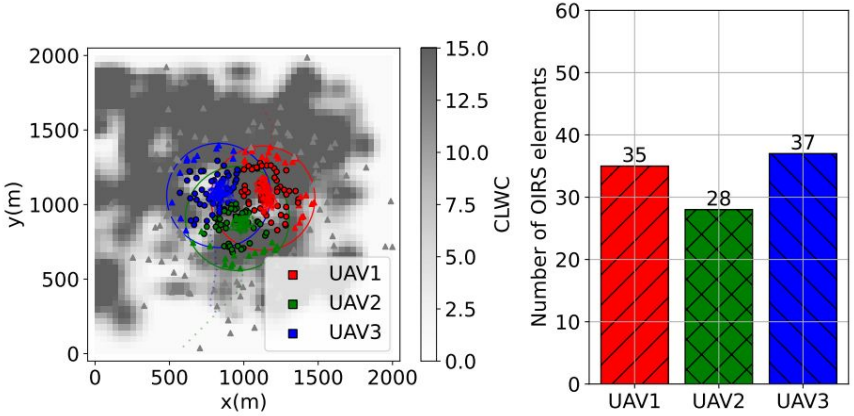
OIRS adaptively allocates the elements according to the number of MUs associated with each UAV and the cloud conditions for each UAV:

Trajectories of UAVs and OIRS element allocation in time slot t=5:



UAV 1 and *UAV 3* need to serve a larger number of MUs.
→ OIRS allocates more elements to them

Trajectories of UAVs and OIRS element allocation in time slot t=1000:

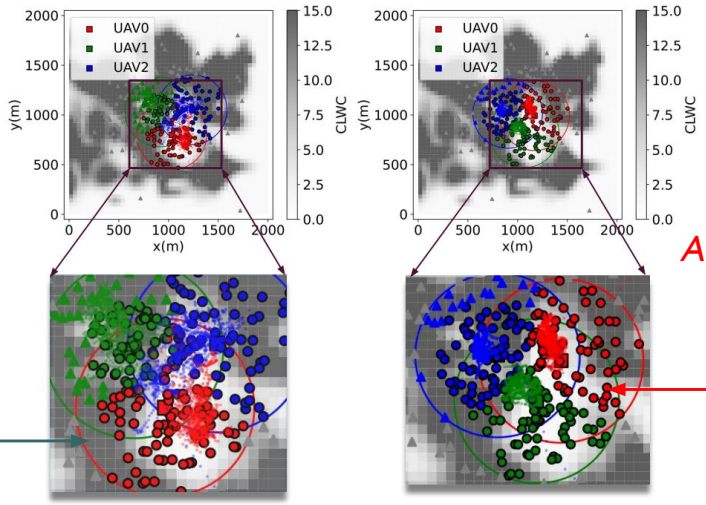


3 UAVs nearly need to serve the same number of MUs and have the same cloud condition.
→ Nearly, OIRS allocates the same number of elements

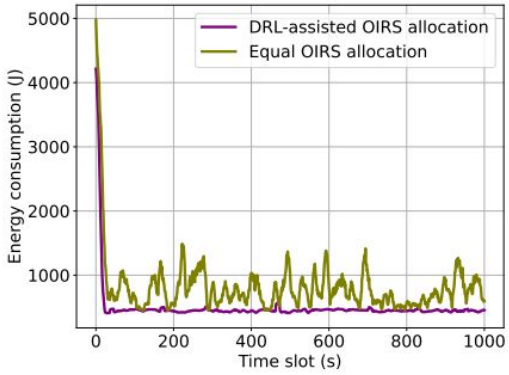
Experimental Results: The Effectiveness of Adaptive OIRS Allocation

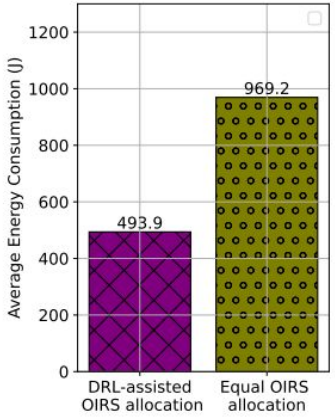
Comparison between equal OIRS allocation and the proposed DRL-assisted OIRS allocation:

UAV Trajectory:

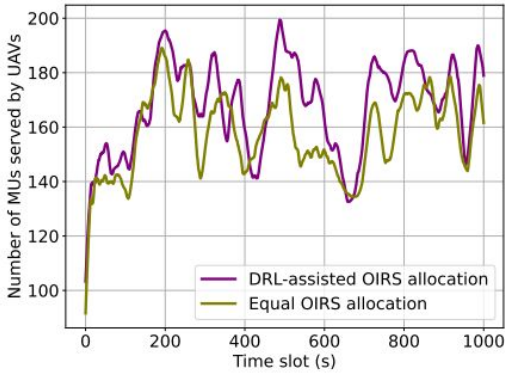


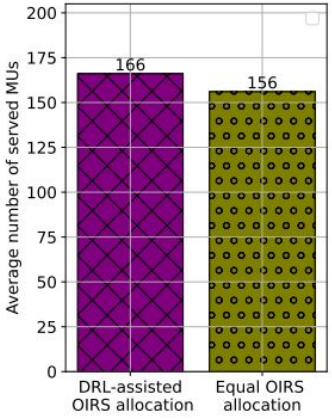
UAV energy consumption:





The number of served MUs:





→ OIRS enhances the number of served MUs and reduces the UAVs' energy consumption

V. Conclusion

Conclusion

- This work introduced a AGIN architecture integrating:
  - An Optical IRS-aided HAP-based FSO backhaul network.
  - UAV-mounted base stations for RF access networks.

→ Deliver high-speed connectivity to remote, rural areas or quickly deployable scenarios e.g., temporary events

- This work proposed a heterogeneous CTDE-based MADRL optimization framework:
  - Propose heterogeneous actors for UAV and OIRS agents
  - Address the saturation problem and invalid action in continuous DRL.

→ Higher reward convergence than DRL benchmarks. Incorporate OIRS agent to enhance the network performance.

Thank you for your listening!